

Vijaykumar Karthikeyan

Los Angeles, CA | (859) 894-0010 | vijayk.karthik15@gmail.com | <https://www.linkedin.com/in/vijaykumar-karthikeyan/>

EDUCATION

University of California, Los Angeles (UCLA) – Bachelor of Science (BS) September 2024 – June 2028

- **Bioengineering** + Technical Breadth in **Computer Science** + Minor in **Data Science Engineering**

Paul Laurence Dunbar High School (Lexington, KY) August 2020 – May 2024
Math Science Technology Center (selective admissions) + 400-hour Capstone Research

- Unweighted GPA – 4.00 Weighted GPA – 4.83 ACT – 35 16 AP Classes

EXPERIENCE

Undergraduate Researcher – UCLA Bioengineering (The Kamariza Lab) January 2025 – **Present**

- Studying the interactions of **LwaCas13a-CRISPR** Assays under the mentorship of *Dr. Mireille Kamariza*
- Predicting LwaCas13a-CRISPR Activation based on molecular interactions and Machine Learning

Cofounder, CTO – Cogenesis, Inc. March 2025 – November 2025

- Incubated at *Founder's, Inc.* and *Startup UCLA*; Developed AI for Microscopic Image Analysis with Meta SAM

Student Researcher – University of Kentucky CAER (The Risko Lab) October 2022 – June 2024

- Developed ML Algorithms to predict the **interactions of Organic Semiconductors (OSC)** under *Dr. Chad Risko*
- Validated use of Active Learning for OSC Research and Presented at *Kentucky Science and Engineering Fair*

HIGHLIGHTED PROJECTS - <https://kvijay3.github.io/>

Ateball – Creative Labs @ UCLA October 2025 – **Present**

- Created Backend Application that uses Google Places API and smart scoring to allow users to pick places to eat

Biomedical Engineering Society (BMES) @ UCLA September 2024 – **Present**

Kinetiq – Knee Brace (2025/26 Design Team)

- Full Stack Development of Doctor and Patient Dashboards to display analytics from a Smart Knee Brace
- Creating Analytics from raw sensor data to track Knee Health and aid in treatment planning

Kwo – iOS Application for ML Health Prediction (2024/25 Design Team)

- Designed **Django Rest Framework** to process Apple HealthKit data and host LLMs on **AWS EC2** instances

VEST @ UCLA January 2025 – **Present**

- Community of the best builders at UCLA, organizing and engaging with VCs/builders across SoCal
- Created AI-Powered content generation to create hook videos for lookbk.com (Shazam for Clothes)

SKILLS

Programming & Data Science – Python, JavaScript, C++, Pytorch, scikit-learn, ML/LLM Inferencing/Finetuning

Backend Engineering & Automation – AWS Cloud (EC2, Lambda, S3), REST API, MongoDB, Docker, FireWorks

Frontend Engineering & Design – Swift UI (macOS, iOS), Flutter, Figma, Canva, MS Office, AWS Amplify

Laboratory Techniques – In Vitro Transcription, QuantStudio 5, PCR, Pipetting, Microscopy Analysis, Benchling

Languages – English (Fluent), German (Limited Proficiency), Tamil (Fluent), Telugu (Fluent)

Interests – Soccer (playing and watching), Biking, Electronics, Learning about RNA technologies